

DATA SCIENTIST

OLUWASEMILORE ADELAJA

I work as a data scientist and Machine learning energy, specialising specifically in Deep Learning.

I am also an aspiring chess programmer

EDUCATION

University of Lagos

Mechanical Engineering | 10 2021 - 07 2027
Lagos, Nigeria

PROJECTS

MR & N

This is my "Math, Research and Numpy" project where I create a Deep Learning Framework with as many functionalities with only basic python and Numpy.

I implemented Fully Connected Layers, Dropout, Momentum based gradient descent, Convolutional Neural Networks, learning rate scheduling, as well as a variety of activation and cost functions.

I focus on modularity, simplicity and efficiency for the framework for the lofty hope of dethroning industry standards in its finished form

Hexapawn

This is my preparatory project before I tackle chess programming.

In this project, I train a Neural Network to be able to play a depressed version of chess known as Hexapawn.

It is a showcase of Reinforcement Learning Ideas as the AI is modelled after AlphaZero in both training and inference. It uses a Neural Network alongside a Monte Carlo Tree Search to chose best moves and play perfectly the game of Hexapawn

Transformer

I implement the network described in the "Attention is all you need" paper.

CONTACT

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LINKS

Github

SKILLS

Python

Scipy

C#

COURSES

Mechanical engineering
University of Lagos
10 2021 - 06 2027

LANGUAGES

English

Yoruba

In this project, I implement a decoder based Neural Network from scratch and evaluate its performance by its ability to recreate shakespearean sounding literature.

AudioNet ESC-50 Classifier

My foray into Audio classification where I implement a Neural Network to classify sounds by converting them into frequency maps and leverage transfer learning from finetuned image recognising Neural Networks.

I evaluate my model on the ESC-50 dataset and use the Efficient-Net family of Networks to successfully classify audio data.

Chess Programming

I implement from scratch a Chess Playing Bot. I start from an efficient representation of the board position as well as a quick move generation algorithm.

I also implement chess playing algorithms and heuristics to develop a strong bot. I use minimax, alpha-beta pruning, quiescent search amongst other classic techniques.

Lecture Kit

In this project I develop a software that transforms recordings of lectures to complete and formatted pdfs that contains the extracted academic information contained within the transcript.

The output contains the raw transcript as well as summary of the lecture and definition of prominent terms contained within the lecture.